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- Introduction
- Bayesian Reasoning
- Bayesian Statistical inference
- Probabilistic Modeling
- Bayesian Computation

Bayesian Statistics and Data Analysis
Lecture 1

Måns Magnusson
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Thanks to Aki Vehtari, Aalto University



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Section 1

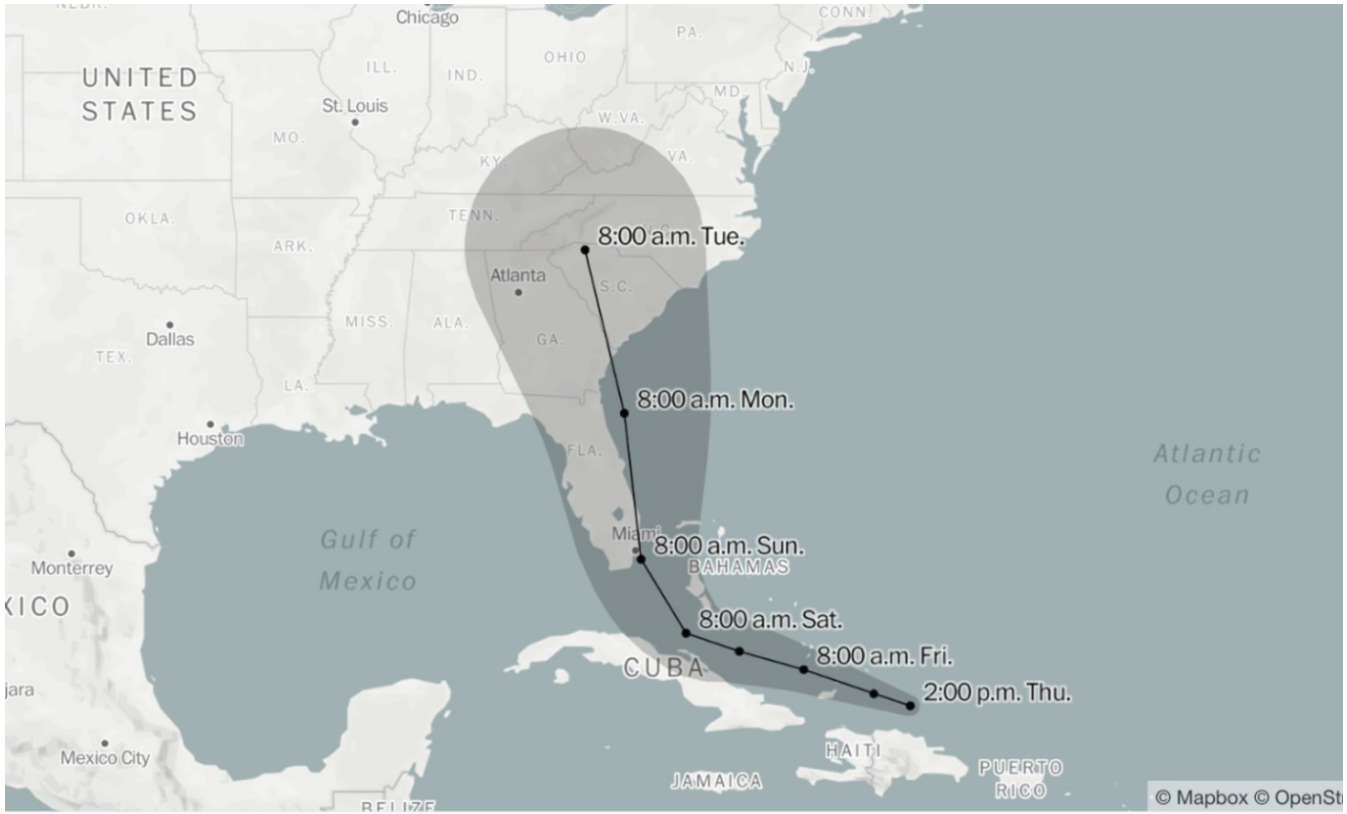
Introduction



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Decision making in case of uncertainty



Tip! See Matthew Kay's StanCon 2026 keynote:
Systematic uncertainty visualization design



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Quick check: uncertainty in decisions

- Think of one decision from your own life, studies, or research where the outcome is uncertain.



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- What are the possible outcomes?
- What information would change your uncertainty?



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Quick check: uncertainty in decisions

- Think of one decision from your own life, studies, or research where the outcome is uncertain.
- What are the possible outcomes?
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- Pair discussion: what would be the analogue of prior information, data, and a decision?



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Bayesian Analysis

- Bayesian probability theory
  - uncertainty is represented with probabilities



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  - uncertainty is represented with probabilities
  - probabilities are updated based on new information
  - *...common sense reduced to calculation*, Laplace 1819



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- Thomas Bayes (170?–1761)
  - English nonconformist, Presbyterian minister, mathematician
  - considered the problem of *inverse probability*



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  - English nonconformist, Presbyterian minister, mathematician
  - considered the problem of *inverse probability*
    - significant part of the Bayesian theory
- Bayes did not invent everything, but was first to solve a problem of inverse probability in a special case
- Modern Bayesian theory with rigorous proofs developed in 20th century



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Term Bayesian used first time in mid 20th century

- Earlier there was just "probability theory"
  - the concept of probability was not strictly defined, although it was close to modern Bayesian interpretation
  - at the end of 19th century there was increasing demand for a stricter definition of probability (mathematical and philosophical problem)



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- In the beginning of 20th century frequentist view gained popularity
  - accepts definition of probabilities only through frequencies
  - does not accept inverse probability or use of prior
  - gained popularity due to apparent objectivity and "cook book" like reference books



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  - term became quickly popular, because alternative descriptions were longer



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  - after this Bayesians started to use term "frequentist"



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Bayesian core: Uncertainty and probabilistic modeling

- Representing uncertainty with probabilities
- Updating uncertainty (with empirical observations)



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Two types of uncertainty

- Aleatoric uncertainty due to randomness
  - observations can't reduce this uncertainty
- Epistemic uncertainty due to lack of knowledge



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Two types of uncertainty

- Aleatoric uncertainty due to randomness
  - observations can't reduce this uncertainty
- Epistemic uncertainty due to lack of knowledge
  - we can reduce this uncertainty with observations
  - two observers may have different epistemic uncertainty



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Quick check: aleatoric or epistemic?

- Classify each uncertainty, then compare with your neighbour:
  1. the next roll of a fair die
  2. the unknown fraction of red chips in a bag
  3. tomorrow's temperature in Uppsala



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Quick check: aleatoric or epistemic?

- Classify each uncertainty, then compare with your neighbour:
  1. the next roll of a fair die
  2. the unknown fraction of red chips in a bag
  3. tomorrow's temperature in Uppsala
- Die roll: mainly aleatoric in this model
- Red-chip fraction: epistemic; sampling chips can reduce it
- Temperature: both; data and forecasts reduce epistemic uncertainty, but weather variation remains



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Section 2

Bayesian Reasoning



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Steve: farmer or librarian?

In a hypothetical population containing only farmers and librarians, Steve is selected uniformly at random.

> *Steve is reserved, comfortable working alone, and attentive to detail.*

- Which occupation seems more likely?
- What probability would you assign to "librarian"?
- What additional information would help?

Adapted from Tversky & Kahneman (1974), *Judgment under Uncertainty: Heuristics and Biases*, *Science*, 185, 1124–1131. The population and all numerical values in this walkthrough are invented for teaching.



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- What probability would you assign to "librarian"?
- What additional information would help?

There is not yet enough information for a unique numerical answer. Keep your initial estimate for later.

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Probability in Bayesian Reasoning

- Uncertainty and knowledge are expressed as probability distributions, why?



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Probability in Bayesian Reasoning

- Uncertainty and knowledge are expressed as probability distributions, why?
- Analogy to physical randomness



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Probability in Bayesian Reasoning

- Uncertainty and knowledge are expressed as probability distributions, why?
- Analogy to physical randomness
- Knowledge as coherence of bets (the Dutch book):
  Example: If you say: P(rain) = 0.7 and P(no rain) = 0.4, those add up to 1.1 (too high).
  A bookmaker could then take both bets from you and be sure to profit, because you have made your odds incoherent.



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Probability in Bayesian Reasoning

- Uncertainty and knowledge are expressed as probability distributions, why?
- Accuracy arguments: incoherent probabilities can be systematically worse estimates of truth
- Conditionalization: update by zeroing possibilities ruled out by evidence and rescaling the rest



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- Everyone can have their own 'subjective' uncertainty, e.g. P(rain tomorrow),
  P(Magdalena Andersson will be the next prime minister)



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- Everyone can have their own 'subjective' uncertainty, e.g. P(rain tomorrow),
  P(Magdalena Andersson will be the next prime minister)
- Frequency arguments can be difficult in some situations: P(other life in the universe)
- See Stanford Encyclopedia of Philosophy: Bayesian Epistemology



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Bayesian epistemology

- What is epistemology?



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Bayesian epistemology

- What is epistemology?
  "the theory of knowledge" (Wikipedia)



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- Bayesian epistemology:



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  "the theory of knowledge" (Wikipedia)
- Bayesian epistemology:
  - Studies rational degrees of belief, or credences



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  - Two core norms:
    - Probabilism: beliefs at one time should obey the rules of probability



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    - Conditionalization: beliefs should be updated by conditioning on new evidence
  - e.g. unlike Popperian approaches, we can talk about P(black swan)
  - Main debates: coherence, priors, idealization, and how evidence should change beliefs
  - Source: Stanford Encyclopedia of Philosophy



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Pascalian and Baconian probability

L. Jonathan Cohen distinguishes two ways of assessing a claim:

- **Pascalian:** How probable is the claim H, given the evidence E? Uses the ordinary probability calculus, for example $P(H \mid E) = 0.8$.
- **Baconian:** How thoroughly has the claim been supported by investigating relevant possibilities and testing alternative explanations?
- In the Baconian sense, little support for a claim need not mean strong support for its opposite: the investigation may simply be incomplete.

In this course, we calculate Pascalian probabilities, but also critically examine the evidence and modelling assumptions behind them.

Source: L. Jonathan Cohen (1977), *The Probable and the Provable*.



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Pair discussion: trusting a probability

Discuss in pairs (1–2 minutes).

An analyst reports:

> *"There is a 90% probability that this batch meets the quality standard."*

- What does this probability tell us?
- What would you want to know about the evidence and assumptions before trusting the conclusion?

Consider how items were sampled, how quality was measured, and whether important sources of failure were investigated.



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Section 3

Bayesian Statistical inference



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Statistical inference

- Draw conclusions about unobserved quantities, based on data



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Statistical inference

- Draw conclusions about unobserved quantities, based on data
- Different types of unobserved quantities
  - potentially observed: future observations, potential outcomes,
  - parameters: data-generating process (e.g. regression coefficients)



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Notation

- $p(\cdot|\cdot)$: conditional pdf/pmf
- $p(\cdot)$: (marginal) pdf/pmf
- $P(\cdot)$ or $\Pr(\cdot)$: probability, e.g. $P(\theta > 0) = \int_0^\infty p(\theta) d\theta$
- Random variable: $\theta \sim \mathcal{N}(\mu, \sigma^2)$
- pdf/pmf: $p(\theta) = \mathcal{N}(\theta|\mu, \sigma^2)$
- We often suppress known covariates/data x and assumptions h



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Parameters, data, and predictions

- Model parameters: $\theta = (\theta_1, ..., \theta_p)$
- Observed data: $y = (y_1, ..., y_n)$
  - y_i can be a vector and is assumed to come from a data generating process (probabilistic model)
- Potentially observed data: $\tilde{y} = (\tilde{y}_1, ..., \tilde{y}_m)$
- Observed (known) covariates: x



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Parameters, data, and predictions

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- Observed data: $y = (y_1, ..., y_n)$
  - y_i can be a vector and is assumed to come from a data generating process (probabilistic model)
- Potentially observed data: $\tilde{y} = (\tilde{y}_1, ..., \tilde{y}_m)$
- Observed (known) covariates: x
- I can be sloppy: e.g. often implicitly condition on x, i.e. $p(\theta|y) = p(\theta|y, x)$



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Quick check: model or likelihood?

- Suppose $Y|\theta \sim \text{Bin}(10, \theta)$, so

$$p(y|\theta) = \binom{10}{y} \theta^y (1-\theta)^{10-y}.$$



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- If $y = 7$ is observed and θ varies, what is this function called?



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- If $y = 7$ is observed and θ varies, what is this function called?
- Is $L(\theta|y)$ a probability distribution over θ?



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Model and likelihood

- Same expression, different role:

$$p(y|\theta)$$



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Model and likelihood

- Same expression, different role:

$$p(y|\theta)$$

- As a function of y with θ fixed: observation model

$$y|\theta \sim p(\cdot|\theta)$$



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$$y|\theta \sim p(\cdot|\theta)$$

- As a function of θ with y fixed: likelihood

$$L(\theta|y) = p(y|\theta)$$



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Model and likelihood

- Same expression, different role:

$$p(y|\theta)$$

- As a function of y with θ fixed: observation model

$$y|\theta \sim p(\cdot|\theta)$$

- As a function of θ with y fixed: likelihood

$$L(\theta|y) = p(y|\theta)$$

- The likelihood is generally not normalized over θ



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The basic steps of Bayesian inference

1. Setting up a full probability model $p(y|\theta) \cdot p(\theta) = p(y, \theta)$



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The basic steps of Bayesian inference

1. Setting up a full probability model $p(y|\theta) \cdot p(\theta) = p(y, \theta)$
2. Conditioning on observed data y to calculate the posterior distribution $p(\theta|y)$



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The basic steps of Bayesian inference

1. Setting up a full probability model $p(y|\theta) \cdot p(\theta) = p(y, \theta)$
2. Conditioning on observed data y to calculate the posterior distribution $p(\theta|y)$
3. Evaluate the model. If not satisfied, go back to 1. Else use the model.



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Bayesian Inference I

- Bayesian conclusions about θ or $\tilde{y}$ are made using probabilities, conditional on data y



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Bayesian Inference I

- Bayesian conclusions about θ or $\tilde{y}$ are made using probabilities, conditional on data y
- We state our uncertainty about θ or $\tilde{y}$ as probability distributions



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Bayesian Inference I

- Bayesian conclusions about θ or $\tilde{y}$ are made using probabilities, conditional on data y
- We state our uncertainty about θ or $\tilde{y}$ as probability distributions
  - parameters: $p(\theta|y)$
  - potentially observed: $p(\tilde{y}|y)$



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Bayesian Inference: Setting up the model

- Start out with a model: a *joint distribution* for data and parameters:

$$p(y, \theta) = p(y|\theta)p(\theta)$$



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Bayesian Inference: Setting up the model

- Start out with a model: a *joint distribution* for data and parameters:

$$p(y, \theta) = p(y|\theta)p(\theta)$$

- $p(y|\theta)$ is our data model, and when conditioned on y, the *likelihood*



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Bayesian Inference: Setting up the model

- Start out with a model: a *joint distribution* for data and parameters:

$$p(y, \theta) = p(y|\theta)p(\theta)$$

- $p(y|\theta)$ is our data model, and when conditioned on y, the *likelihood*
- $p(\theta)$ is our prior distribution for our parameters



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Bayesian Inference: Computing the posterior

- Conditioning on data y, using *Bayes' theorem*, we can compute the *posterior distribution*

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

where

$$p(y) = \int p(y|\theta)p(\theta)d\theta$$



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- $p(y)$ is the *marginal likelihood*



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$$p(y) = \int p(y|\theta)p(\theta)d\theta$$

- $p(y)$ is the *marginal likelihood*
- $p(\theta|y)$ summarize our knowledge about θ



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- $p(y)$ is the *marginal likelihood*
- $p(\theta|y)$ summarize our knowledge about θ
- Bayesian statistics obey the *likelihood principle*: data only affects $p(\theta|y)$ through the likelihood $p(y|\theta)$



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Steve: reveal the base rates

Return to Steve, selected uniformly from our hypothetical population:

- 100 librarians (L) and 2,000 farmers (F)
- Description D: reserved, comfortable working alone, attentive to detail

Before using the description, the base-rate prior is

$$P(L) = \frac{100}{2100} = \frac{1}{21} \approx 4.8\%.$$

Does this settle the question, or do we still need something?



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Does this settle the question, or do we still need something?

We need to know how common the description is in each occupation: $P(D \mid L)$ and $P(D \mid F)$.

The 4.8% is a baseline, not the posterior after considering D.



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Steve: reveal the likelihoods

Suppose the description D matches the following people in our hypothetical population:

Occupation	Total	Matching D
Librarian	100	80 (80%)
Farmer	2,000	200 (10%)

Discuss in pairs (1–2 minutes). Among people matching D, what proportion are librarians?



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Steve: reveal the likelihoods

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Occupation	Total	Matching D
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Farmer	2,000	200 (10%)

Discuss in pairs (1–2 minutes). Among people matching D, what proportion are librarians?

$$P(L \mid D) = \frac{80}{80 + 200} = \frac{2}{7} \approx 28.6\%.$$

The description raises the probability of librarian from 4.8% to 28.6%, but farmer remains more probable.



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Steve: from counts to Bayes' theorem

- **Prior:** $P(L) = 1/21$, $P(F) = 20/21$
- **Likelihoods:** $P(D \mid L) = 0.8$, $P(D \mid F) = 0.1$

$$P(L \mid D) = \frac{P(D \mid L)P(L)}{P(D \mid L)P(L) + P(D \mid F)P(F)}$$

$$= \frac{0.8 \cdot (1/21)}{0.8 \cdot (1/21) + 0.1 \cdot (20/21)} = \frac{2}{7}.$$

The description is eight times as likely among librarians, but there are twenty times as many farmers.

Evidence favouring a hypothesis need not make it the most probable hypothesis.

Representativeness: judging probability by resemblance to a stereotype can neglect the base rates.



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Steve: one more observation

Now we learn that Steve works mainly indoors (I).

Among people already matching the description D, suppose:

Occupation	Also working mainly indoors
Librarian	75% of 80: $80 \cdot 0.75 = 60$
Farmer	10% of 200: $200 \cdot 0.10 = 20$

What is the updated probability that Steve is a librarian?

These invented rates are conditional on D: $P(I \mid L, D) = 0.75$, $P(I \mid F, D) = 0.10$. We have not assumed that D and I are conditionally independent.



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Steve: one more observation

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Among people already matching the description D, suppose:

Occupation	Also working mainly indoors
Librarian	75% of 80: $80 \cdot 0.75 = 60$
Farmer	10% of 200: $200 \cdot 0.10 = 20$

What is the updated probability that Steve is a librarian?

$$P(L \mid D, I) = \frac{60}{60 + 20} = 75\%.$$

Librarian is now more probable. The first posterior becomes the prior for this update.

These invented rates are conditional on D: $P(I \mid L, D) = 0.75$, $P(I \mid F, D) = 0.10$. We have not assumed that D and I are conditionally independent.



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Pair discussion: Steve's evidence

Discuss in pairs (1–2 minutes).

- What if Steve was actually selected at a library conference?
- What would we need to check about the description-matching percentages before using them in a real analysis?
- What if the description was unreliable?



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Pair discussion: Steve's evidence

Discuss in pairs (1–2 minutes).

- What if Steve was actually selected at a library conference?
- What would we need to check about the description-matching percentages before using them in a real analysis?
- What if the description was unreliable?

Pascalian calculation: the probabilities follow from our stated model and evidence.

Baconian concern: how thoroughly have the evidence and alternative explanations been investigated?

Bayesian analysis can address these concerns too: revisit the sampling process, model measurement uncertainty, and check assumptions.



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Predictions

- How to do inference on an unknown (potential) observable $\tilde{y}$? E.g. a future observation, potential outcome



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Predictions

- How to do inference on an unknown (potential) observable $\tilde{y}$? E.g. a future observation, potential outcome
- We use our *data* model and our posterior and *marginalize* over the uncertainty

$$p(\tilde{y}|y) = \int p(\tilde{y}|\theta)p(\theta|y)d\theta$$

- The *posterior predictive distribution*
  'An average of conditional predictions over the posterior distribution'



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Quick check: updating with chips

- A bag has an unknown fraction θ of red chips.
- Before computing anything, predict what happens to $p(\theta|y)$ after:
  1. one red chip
  2. red, red, red
  3. red, yellow, red, yellow



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Quick check: updating with chips

- A bag has an unknown fraction θ of red chips.
- Before computing anything, predict what happens to $p(\theta|y)$ after:
  1. one red chip
  2. red, red, red
  3. red, yellow, red, yellow
- Red observations move the posterior toward larger θ.
- More observations usually make the posterior more concentrated.
- Mixed observations keep the posterior closer to intermediate values.



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Example: Updating uncertainty



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Example: Updating uncertainty

- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$



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Example: Updating uncertainty

- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$
- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty



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Example: Updating uncertainty

- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$
- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty
- $p(\theta)$ epistemic uncertainty



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Example: Updating uncertainty

- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$
- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty
- $p(\theta)$ epistemic uncertainty
- Picking many chips updates our uncertainty about the proportion
- $p(\theta|y_1 = \text{red}, y_2 = \text{yellow}, y_3 = \text{red}, \ldots) =?$



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Example: Updating uncertainty

- Probability of red $\frac{\#\text{red}}{\#\text{red}+\#\text{yellow}} = \theta$
- $p(y = \text{red}|\theta) = \theta$ aleatoric uncertainty
- $p(\theta)$ epistemic uncertainty
- Picking many chips updates our uncertainty about the proportion
- $p(\theta|y_1 = \text{red}, y_2 = \text{yellow}, y_3 = \text{red}, \ldots) = ?$
- Bayes rule $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$



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Model vs. Likelihood

- Bayes rule

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} \propto p(y|\theta)p(\theta)$$

- Data model: $p(y|\theta)$ as a function of y given fixed θ describes the aleatoric uncertainty

- Likelihood: $p(y|\theta) = L(\theta|y)$ as a function of θ given fixed y provides information about epistemic uncertainty.but is not a probability distribution.



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Model vs. Likelihood

- Bayes rule

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} \propto p(y|\theta)p(\theta)$$

- Data model: $p(y|\theta)$ as a function of y given fixed θ describes the aleatoric uncertainty
- Likelihood: $p(y|\theta) = L(\theta|y)$ as a function of θ given fixed y provides information about epistemic uncertainty.but is not a probability distribution.
- Bayes rule combines the likelihood with prior uncertainty $p(\theta)$ and transforms them to updated posterior uncertainty $p(\theta|y)$



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Example application: Effects on roaches

- Urban pest-management experiment in apartments
- Question: Does treatment reduce the number of captured roaches?



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Example application: Effects on roaches

- Urban pest-management experiment in apartments
- Question: Does treatment reduce the number of captured roaches?
- Outcome (y_i): The number of roaches captured in home i
- Covariates (x_i): Treatment, senior home, pre-treatment number of roaches
- Exposure (u_i): Different apartments used traps for different numbers of days
- Source: Gelman and Hill (2007, Ch. 8.3); Vehtari roaches cross-validation demo
- Parameters (θ): Intercept α and coefficients β



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Example application: Effects on roaches

- Urban pest-management experiment in apartments
- Question: Does treatment reduce the number of captured roaches?
- Outcome (y_i): The number of roaches captured in home i
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- Exposure (u_i): Different apartments used traps for different numbers of days
- Source: Gelman and Hill (2007, Ch. 8.3); Vehtari roaches cross-validation demo
- Parameters (θ): Intercept α and coefficients β
- Data model:

$$y_i|\theta, x_i \sim \text{Poisson}(\lambda_i), \quad p(y_i|\theta, x_i) = \frac{\lambda_i^{y_i} \exp(-\lambda_i)}{y_i!}$$

where

$$\lambda_i = u_i \exp(\alpha + x_i^{\mathsf{T}} \beta)$$



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Example application: Effects on roaches

- Prior:

$$p(\theta) = p(\alpha) \prod_{k=1}^{K} p(\beta_k),$$

$$p(\alpha) = \mathcal{N}(\alpha | \mu_\alpha, \sigma_\alpha^2),$$

$$p(\beta_k) = \mathcal{N}(\beta_k | 0, \sigma_\beta^2)$$



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Example application: Effects on roaches

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$$p(\theta) = p(\alpha) \prod_{k=1}^{K} p(\beta_k),$$

$$p(\alpha) = \mathcal{N}(\alpha | \mu_\alpha, \sigma_\alpha^2),$$

$$p(\beta_k) = \mathcal{N}(\beta_k | 0, \sigma_\beta^2)$$

- Posterior:

$$p(\alpha, \beta | y, X, u) \propto p(\alpha) \prod_{k=1}^{K} p(\beta_k)$$

$$\prod_{i=1}^{N} \text{Poisson}(y_i | u_i \exp(\alpha + x_i^\mathsf{T} \beta))$$



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Example application: Effects on roaches

- Prior:

$$p(\theta) = p(\alpha) \prod_{k=1}^{K} p(\beta_k),$$

$$p(\alpha) = \mathcal{N}(\alpha|\mu_\alpha, \sigma_\alpha^2),$$

$$p(\beta_k) = \mathcal{N}(\beta_k|0, \sigma_\beta^2)$$

- Posterior:

$$p(\alpha, \beta|y, X, u) \propto p(\alpha) \prod_{k=1}^{K} p(\beta_k)$$

$$\prod_{i=1}^{N} \text{Poisson}(y_i|u_i \exp(\alpha + x_i^\mathsf{T}\beta))$$

- Predictive distribution:

$$p(\tilde{y}|\tilde{x}, \tilde{u}, y, X, u) = \int \text{Poisson}(\tilde{y}|\tilde{u} \exp(\alpha + \tilde{x}^\mathsf{T}\beta))$$

$$p(\alpha, \beta|y, X, u)\, d\alpha\, d\beta$$



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Section 4

Probabilistic Modeling



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The art of probabilistic modeling

- Subjectivity: we need to specify both $p(\theta)$ and $p(y|\theta)$
- The art of probabilistic modeling is to describe in a mathematical form (model and prior distributions) what we already know and what we don't know



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The art of probabilistic modeling

- Subjectivity: we need to specify both $p(\theta)$ and $p(y|\theta)$
- The art of probabilistic modeling is to describe in a mathematical form (model and prior distributions) what we already know and what we don't know

- "Easy" part is to use Bayes rule to update the uncertainties
  - computational challenges



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The art of probabilistic modeling

- Subjectivity: we need to specify both $p(\theta)$ and $p(y|\theta)$
- The art of probabilistic modeling is to describe in a mathematical form (model and prior distributions) what we already know and what we don't know

- "Easy" part is to use Bayes rule to update the uncertainties
  - computational challenges

- Other parts of the art of probabilistic modeling are, for example,
  - model checking/assessment: is data in conflict with our prior knowledge?
  - model choice: which model should we use?
  - presentation: presenting and visualising the model and the results to the application experts



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Example applications

- Galaxy clusters for cosmology
- Coagulation of blood
- Gene regulation
- Pharmacokinetics and -dynamics
- Decision support
- Effects of nutrition for diabetes
- Evolutionary anthropology
- Clinical trial designs
- Daily demand for gas
- Brain structure trees
- School enrollment
- Sports
- Product demand
- Cocoa bean fermentation
- Marine propulsion power
- Alcohol consumption trends
- Flood probability
- Instantaneous heart rate distributions
- Drug dosing regimens in pediatrics
- Human T stem cell memory cells
- Fairness in university admission policies
- Destruction of bacteria and bacterial spores under heat



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Benefits and problems when going Bayesian

- Benefits of Bayesian approach
  - uncertainty and probability theory is the foundation
  - easier interpretation of uncertainty intervals
  - integrate over uncertainties to focus to interesting parts
  - straightforward predictive distributions
  - use relevant prior information
  - straightforward hierarchical models
  - straightforward model checking and evaluation
  - easier to build and evaluate very complex models



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Benefits and problems when going Bayesian

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  - uncertainty and probability theory is the foundation
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  - integrate over uncertainties to focus to interesting parts
  - straightforward predictive distributions
  - use relevant prior information
  - straightforward hierarchical models
  - straightforward model checking and evaluation
  - easier to build and evaluate very complex models
- Complications of Bayesian approach
  - most models do not have nice analytical posteriors
  - we need to *approximate* our posterior
  - can be computationally costly/take long time to estimate complex models



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Section 5

Bayesian Computation



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Computation

We want to compute expectations with respect to a posterior distribution $p(\theta|y)$

$$\mathrm{E}_{\theta|y}\left[g(\theta)\right] = \int p(\theta|y)g(\theta)d\theta$$

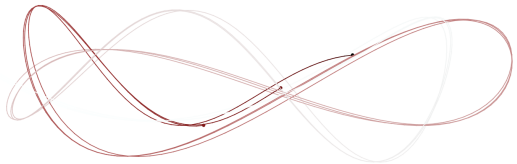
- Analytic
  - only for very simple models
- Monte Carlo, Markov chain Monte Carlo
  - generic, high-quality
  - computationally expensive
- Distributional approximations
  - e.g. Laplace, variational inference
  - less generic, but can be much faster with sufficient accuracy



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Probabilistic programming



Enables agile workflow for developing probabilistic models

language – automated inference – diagnostics



mc-stan.org



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Binomial model for treatment/control comparison

- Two groups of patients: treatment and control
  - Binary outcome (1 success, 0 failure)
  - Is the treatment useful?
- Model:
  Likelihood

$$y_0|\theta_0, N_0 \sim \text{Bin}(N_0, \theta_0)$$
$$y_1|\theta_1, N_1 \sim \text{Bin}(N_1, \theta_1)$$



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$$y_1|\theta_1, N_1 \sim \text{Bin}(N_1, \theta_1)$$

  Prior

$$\theta_0 \sim \text{Beta}(1, 1)$$
$$\theta_1 \sim \text{Beta}(1, 1)$$



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  Prior

$$\theta_0 \sim \text{Beta}(1, 1)$$
$$\theta_1 \sim \text{Beta}(1, 1)$$

  Posterior

$$p(\theta_0, \theta_1|y_0, y_1) \propto \prod_{g=0}^{1} \text{Bin}(y_g|N_g, \theta_g)\,\text{Beta}(\theta_g|1, 1)$$



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Binomial model for treatment/control comparison

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  Prior

$$\theta_0 \sim \text{Beta}(1, 1)$$
$$\theta_1 \sim \text{Beta}(1, 1)$$

  Posterior

$$p(\theta_0, \theta_1|y_0, y_1) \propto \prod_{g=0}^{1} \text{Bin}(y_g|N_g, \theta_g)\,\text{Beta}(\theta_g|1, 1)$$

  Goal (odds ratio), e.g. $P(OR > 1)$

$$OR = \frac{\frac{\theta_1}{1-\theta_1}}{\frac{\theta_0}{1-\theta_0}}$$



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Binomial model for treatment/control comparison

```
data {
  int<lower=0> N1;
  int<lower=0> y1;
  int<lower=0> N0;
  int<lower=0> y0;
}
parameters {
  real<lower=0,upper=1> theta1;
  real<lower=0,upper=1> theta0;
}
model {
  theta1 ~ beta(1,1);
  theta0 ~ beta(1,1);
  y1 ~ binomial(N1, theta1);
  y0 ~ binomial(N0, theta0);
}
generated quantities {
  real oddsratio;
  oddsratio = (theta1/(1-theta1))/(theta0/(1-theta0));
}
```



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Binomial model for treatment/control comparison

RStanARM

```
fit_bin2 <- stan_glm(cbind(y, N - y) ~ grp2,
                     family = binomial(), data = d_bin2)
```



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Modeling nature

- Drop a ball from different heights and measure time (data)



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Modeling nature

- Drop a ball from different heights and measure time (data)
  - Newton
  - air resistance, air pressure, shape and surface structure of the ball
  - relativity



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Modeling nature

- Drop a ball from different heights and measure time (data)
  - Newton
  - air resistance, air pressure, shape and surface structure of the ball
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- Taking into account the accuracy of the measurements, how accurate model is needed?



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Modeling nature

- Drop a ball from different heights and measure time (data)
  - Newton
  - air resistance, air pressure, shape and surface structure of the ball
  - relativity
- Taking into account the accuracy of the measurements, how accurate model is needed?
  - often simple models are adequate and useful
  - *All models are wrong, but some of them are useful*, George P. Box



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Recap: Uncertainty and probabilistic modeling

- Two types of uncertainty: aleatoric and epistemic
- Representing uncertainty with probabilities
- Updating uncertainty using data



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Rest of the course

- Basic models which can be used as building blocks for complex ones
- Basic computation of posterior and predictive distributions
- Typical simple scientific data analysis cases
  - e.g. comparison of treatments
- Presentation of the results



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Week 1 reading

- BDA3 Chapter 1
  - 1.1–1.3: steps, notation, Bayes' rule
  - 1.4: discrete examples
  - 1.5: probability as uncertainty
  - 1.6: calibration example
  - 1.8: probability review for assignments
- Focus on the terms in the reading instructions